

Software Requirements Specification (SRS)

QR-based Emergency and Parking Contact System

Version 1.0

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1 Introduction

1.1 Purpose

The software is a QR-based emergency and parking contact system. It connects bystanders to a vehicle owner's trusted network without revealing the owner's real phone number, while providing critical medical information.

1.2 Scope

- **Goals:** Contact family in emergencies, resolve parking issues, prevent fraud via AI/GPS checks, and provide instant medical profiles.
- **Benefits:** Privacy (masked numbers), life-saving medical data visibility, and offline SMS fallback.
- **Limits:** Does not include in-app ambulance/police dialing, theft tracking, or physical confrontation features.

1.3 Intended Audience

Developers, testers, clients.

1.4 Definitions & Acronyms

QR Quick Response (code).

GPS Global Positioning System.

AI Artificial Intelligence.

PTA Pakistan Telecommunication Authority.

API Application Programming Interface.

Number Masking A telecom feature that connects two parties via a proxy number without revealing their actual phone numbers to each other.

SMS Fallback An offline text-message trigger (HELP [Reg #] to 8888) used to initiate the contact flow without internet.

2 Overall Description

2.1 Product Perspective

A standalone web-based system for bystanders (requiring no app download) and a back-end/dashboard system for vehicle owners and system admins.

2.2 User Classes and Characteristics

- **Vehicle Owner:** Registers vehicle, medical, and emergency contacts.
- **Bystander/Scanner:** Scans QR or sends SMS to report issues. Requires basic mobile phone skills.
- **Emergency Contact:** Receives masked calls/SMS.
- **System Admin:** Reviews flagged fraud and manages user bans.

2.3 Operating Environment

- **Hardware:** Mobile phone with camera and GPS (for QR scan), basic mobile phone (for SMS fallback), cloud servers.
- **Software:** Mobile web browser, backend server.
- **OS:** Any OS supporting modern mobile web browsers and standard cellular SMS.

2.4 Design and Implementation Constraints

Must use PTA-compliant local telecom APIs (e.g., SendPK/MoceanAPI) for masked calls. Must use YOLOv8/YOLO11m for AI damage detection. Must encrypt medical data at rest.

2.5 Assumptions and Dependencies

Bystanders have access to a mobile phone with a camera or SMS capability. Telecom APIs remain operational and PTA-compliant. AI model accurately detects vehicle damage.

3 System Features and Requirements

3.1 Functional Requirements

- Owner inputs vehicle details, medical profile (with mandatory self-reported disclaimer checkbox), and primary/secondary emergency contacts.
- System generates a unique QR code for the specific vehicle.
- Bystander scans the QR code.
- System displays the medical profile and the self-reported disclaimer.
- System explicitly displays a warning to the bystander that their location (GPS) and timestamp are being tracked and recorded to prevent fraud.
- Bystander selects the issue type (Accident/Parking) and takes a live photo (gallery upload disabled).
- System captures the bystander's GPS coordinates and timestamp.

- System runs parallel fraud checks: Time (< 15 mins), GPS match, and AI damage detection.
- System triggers auto-cascade: Calls Primary (30s) → Calls Secondary (30s) → Sends SMS/WhatsApp to both.
- System restricts the scan session to 30 minutes or a maximum of 3 calls.
- Bystander sends SMS HELP [Vehicle Reg #] to 8888 (offline fallback).
- System receives the SMS via telecom API, parses the registration number, looks up the owner, and triggers the auto-cascade calling sequence without requiring a photo.
- System enforces daily call caps, tracks cross-day harassment patterns (warning → temp ban → perm ban), displays owner scan logs, and generates a police evidence bundle.

3.2 External Interfaces

TBD

3.3 Non-Functional Requirements

- **Speed:** Web page must load instantly for the bystander.
- **Safety:** Medical data must be encrypted at rest. Real phone numbers must never be revealed (Number Masking).
- **Reliability:** Auto-cascade must execute automatically without requiring the bystander to re-scan or intervene.

4 Appendices

4.1 Glossary

Auto-Cascade Automated sequential calling of emergency contacts.

Triple Lock The fraud check combining Time, GPS, and AI damage detection.

Session The temporary state created after one QR scan, limiting calls/time.

Number Masking Hiding real phone numbers during a call using a proxy.

SMS Fallback The offline SMS method to trigger alerts without internet.

4.2 Analysis Models

TBD